CDA 106
Numerical Methods for Data Science

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Ex find $\int_0^2 \frac{\pi}{4} x^4 \cos\frac{\pi x}{4}\, dx = ?$

Simpson's $\frac{1}{3}$rd rule

Ans) $n = 2$

$a = 0$, $b = 2$, $f(x) = \frac{\pi}{4} x^4 \cos\frac{\pi x}{4}$ $\qquad h = \frac{2-0}{2} = 1$

$x_0 = 0$, $x_1 = 1$, $x_2 = 2$

$f(x_0 = 0) = 0$ $\qquad f(x_1) = \frac{\pi}{4} \cdot \frac{1}{\sqrt{2}}$ $\qquad f(x_2) = 0$

f_0 $\qquad f_1$ $\qquad f_2$

$$\int_0^2 f(x)\,dx = \frac{h}{3}\left[f_0 + 4f_1 + f_2\right] = \frac{1}{3}\left[0 + \frac{4\pi}{4\sqrt{2}} + 0\right] = 0.7404$$

Ans.

Ans 2) $n = 4$

$h = \frac{2-0}{4} = 0.5$

$x_j = jh, \; 0 \le j \le 4$

$f(x) = \frac{\pi}{4} x^4 \cos\frac{\pi x}{4}$

j	x_j	$f(x_j)$
0	0	0 $\rightarrow f_0$
1	0.5	0.045351 $\rightarrow f_1$
2	1	0.555 36 $\rightarrow f_2$
3	1.5	1.525579 $\rightarrow f_3$
4	2	0 $\rightarrow f_4$

$x_0 \quad x_1 \quad x_2 \quad x_3 \quad x_4$

$f_0 \quad f_1 \quad f_2 \quad f_3 \quad f_4$

$$\int_0^2 f(x)\,dx = \frac{h}{3}\left[f_0 + 2\times f_2 + 4(f_1+f_3) + f_4\right]$$

$$= 1.22974$$

Important $\rightarrow$ We are getting better value.

Find $\cos(1.5) = 0.070737$ Correct

0.99965 Wrong.

$\cos(1.5^\circ)$

$\rightarrow \cos(x)$ $x \rightarrow$ Radian

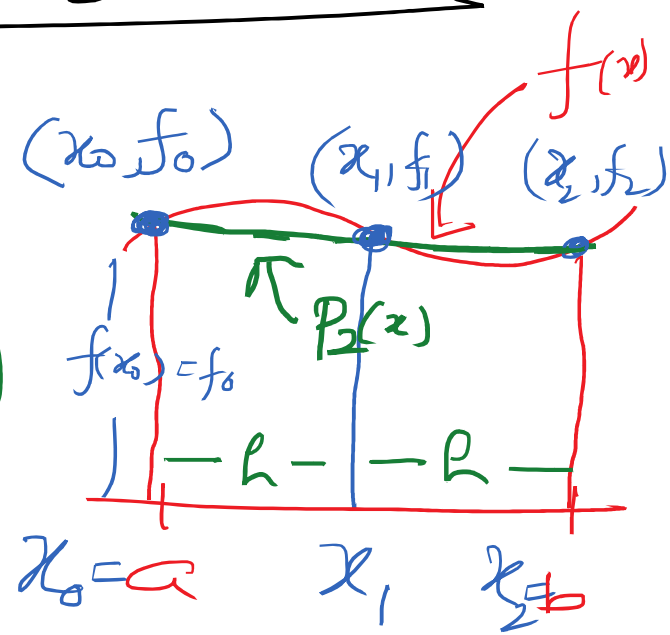
(NOT in Degree)

$360^\circ = 2\pi$ Radian

Derivation of Simpson's $\frac{1}{3}$rd rule

Quadratic Eq^n : $Q(x) = Ax^2 + Bx + C$
passing through (x_0, f_0), (x_1, f_1) & (x_2, f_2).

$$h = \frac{b-a}{2}$$



$$P_2(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f_2$$

$x = x_0 \quad P_2(x_0) = f_0$

$x = x_1 \Rightarrow P_2(x_1) = f_1$

$x = x_2 \quad P_2(x_2) = f_2$

Lagrange's Quadratic Polynomial

According to Simpson's $\frac{1}{3}$rd rule

$$\int_a^b f(x)\,dx = \int_a^b P_2(x)\,dx$$

(Approximating the function by a Quadratic polynomial)

Transformation: $x_1 - x_0 = h$, $x_2 - x_1 = h$

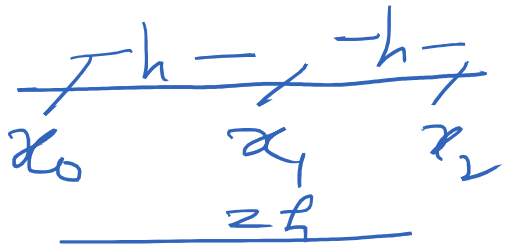
$$\frac{x - x_1}{h} = s \Rightarrow x = x_1 + sh \Rightarrow dx = h\,ds$$

$x - x_1 = sh$ || $x - x_0 = (x - x_1) + h = (s+1)h$

$x - x_2 = (x - x_1) - h = (s-1)h$

$$P_2(x) = \frac{\overbrace{(x-x_1)}^{sh}\overbrace{(x-x_2)}^{(s-1)h}}{\underbrace{(x_0-x_1)}_{-h}\underbrace{(x_0-x_2)}_{-2h}} f_0 + \frac{\overbrace{(x-x_0)}^{(s+1)h}\overbrace{(x-x_2)}^{(s-1)h}}{\underbrace{(x_1-x_0)}_{h}\underbrace{(x_1-x_2)}_{-h}} f_1 + \frac{\overbrace{(x-x_0)}^{(s+1)h}\overbrace{(x-x_1)}^{sh}}{\underbrace{(x_2-x_0)}_{2h}\underbrace{(x_2-x_1)}_{h}} f_2$$

$$= \frac{1}{2}s(s-1)f_0 - (s^2-1)f_1 + \frac{1}{2}s(s+1)f_2$$



When $x = x_0 = a$

$s = -1$

$x = x_2 = b$ $\quad s = 1$

$$\int_a^b P_2(x)\,dx = \int_{-1}^{1} [\quad] \, h\, ds$$

$$\int_a^b p_2(x)\,dx = h\int_{-1}^{1}\left[\left(\frac{s^2}{2}-\frac{s}{2}\right)f_0 - s^2 f_1 + f_1 + \left(\frac{s^2}{2}+\frac{s}{2}\right)f_2\right]ds$$

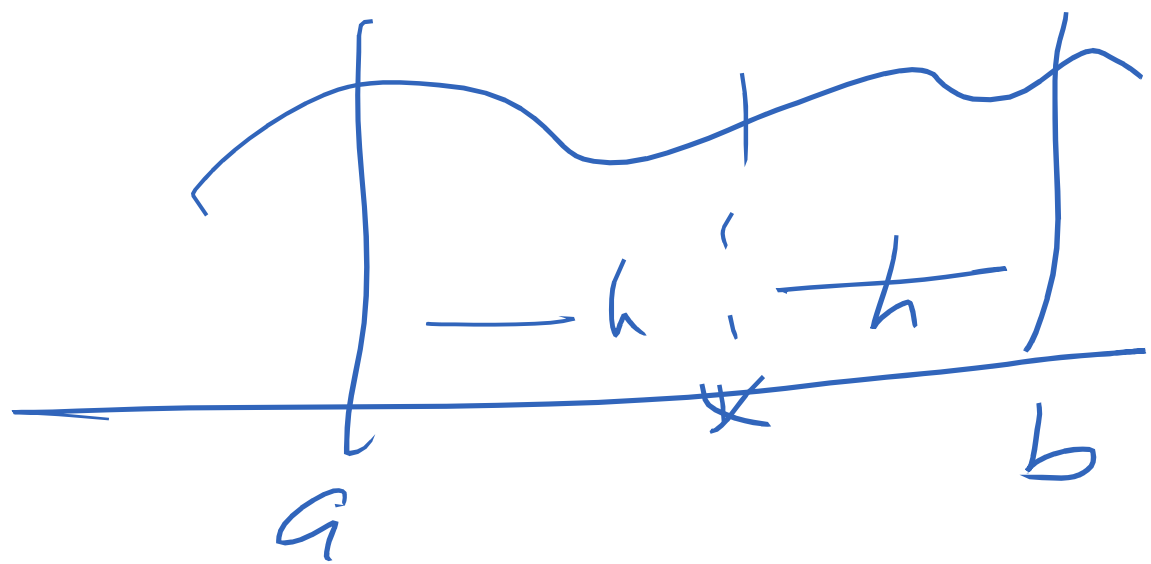
$$= 2h\left[\frac{s^3}{3\times 2}f_0 - \frac{s^3}{3}f_1 + s f_1 + \frac{s^3}{3\times 2}f_2\right]_{s=0}^{s=1}$$

$$= 2h\left[\frac{f_0}{6} + \frac{2}{3}f_1 + \frac{f_2}{6}\right]$$

$$= \frac{1}{3}h\left[f_0 + 4f_1 + f_2\right]$$

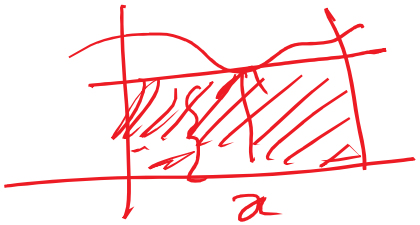
$$\boxed{\int_a^b f(x)\,dx = \frac{h}{3}\left[f_0 + 4f_1 + f_2\right]}$$

$\underline{\underline{n = 2}}$



$$\int_a^b f(x)\,dx = \int_a^b f(x^*)\,dx \qquad \text{(Rectangular Rule)}$$

$$= f(x^*) \times (b-a)$$



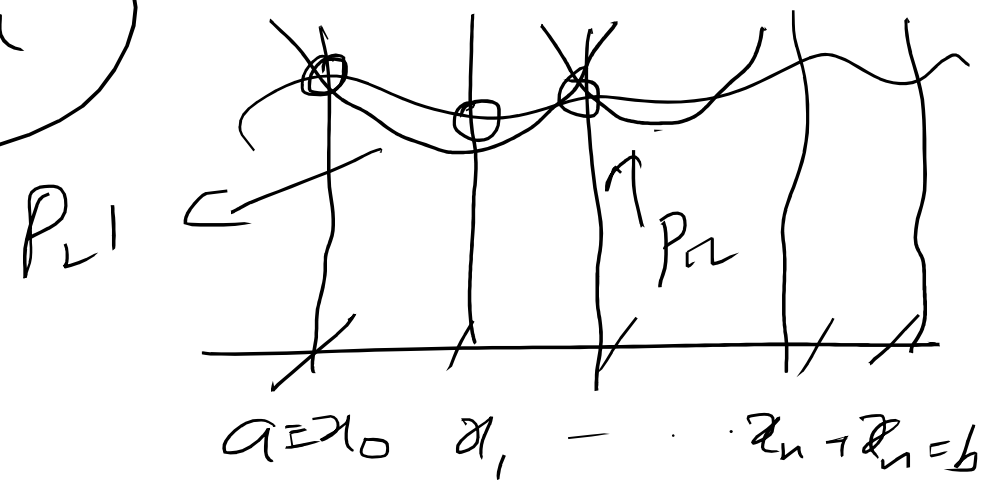
$$\int_a^b f(x)\,dx = \frac{f(a)+f(b)}{2} \times (b-a) \qquad \text{Trap.}$$

$\rightarrow$ Linear eqn

$$\int_a^b f(x)\,dx = \int_a^b (\text{Quadratic eqn})\,dx$$

$$\int_a^b f(x)\,dx$$

n even



$$= \int_{x_0}^{x_2} P_{21}(x)\,dx + \int_{x_2}^{x_4} P_{22}(x)\,dx + \cdots + \int_{x_{n-2}}^{x_n} P_{2n}(x)\,dx$$

$$= \frac{h}{3}\left[f_0 + 4f_1 + f_2\right] + \frac{h}{3}\left[f_2 + 4f_3 + f_4\right] + \cdots$$

$\longrightarrow$ Simpson's 1/3rd formula.

4) Simpson's $\frac{3}{8}$th rule

→ Rectangular $f(x) = \text{const}$

→ Trapezoidal $f(x) = \text{Linear}$

→ Simpson's $\frac{1}{3}$rd $f(x) = \text{quadratic}$

→ Simpson's $\frac{3}{8}$th rule $f(x) = \text{cubic}$

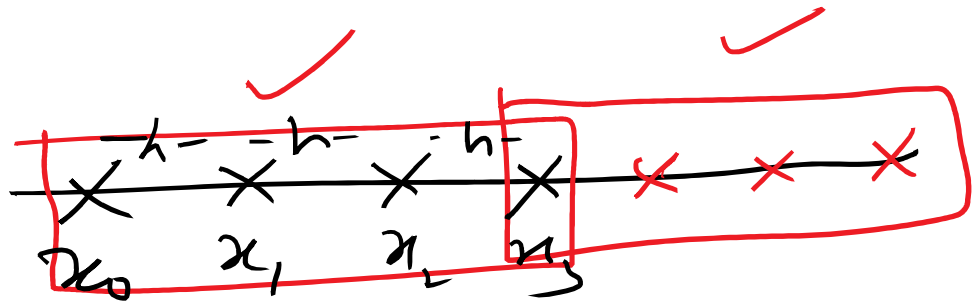
No of partitions $\boxed{n = 3m}$

Apply Simpson's $\frac{1}{3}$rd rule

$[a, b]$ cubic

4 points

3 partitions



$1+3(4)+3+3+\cdots+3$

$3k+1$ points

$n = 3 * \text{multiple}$

partition $= 3k$

Simpson's $\frac{1}{3}$rd Rule $\longrightarrow$ Partitions $n = 2m$

Simpsons $\frac{3}{8}$th rule $\longrightarrow$ $\underline{n = 3m}$

<u>Important</u>

Simpson's $\frac{3}{8}$ Rule

$$\boxed{n = 3m}$$

$$\int_a^b f(x)\,dx = \frac{3h}{8}\Big[f_0 + 3(f_1 + f_2 + f_4 + f_5 + \cdots + f_{n-1} + f_{n-2})$$

$$+ 2(f_3 + f_6 + f_9 + \cdots + f_{n-3}) + f_n\Big]$$

Ex $I = \int_0^1 \frac{1}{1+x^2}\,dx = \frac{\pi}{4} = 0.7853$

To evaluate $\int_0^1 \frac{1}{1+x^2}\,dx = ?$ $\quad n=6$

$a=0 \qquad b=1 \qquad f(x) = 1/1+x^2$

$h = \frac{1-0}{6} = \frac{1}{6} \qquad x_j = jh \quad 0 \le j \le n$

j	x_j	$f_j = f(x_j)$
0	0	1 — f_0
1	1/6	0.973 — f_1
2	1/3	0.9 — f_2
3	1/2	0.8 — f_3
4	2/3	0.746 — f_4
5	5/6	0.64 — f_5
6	1	0.5 — f_6

3/8th rule

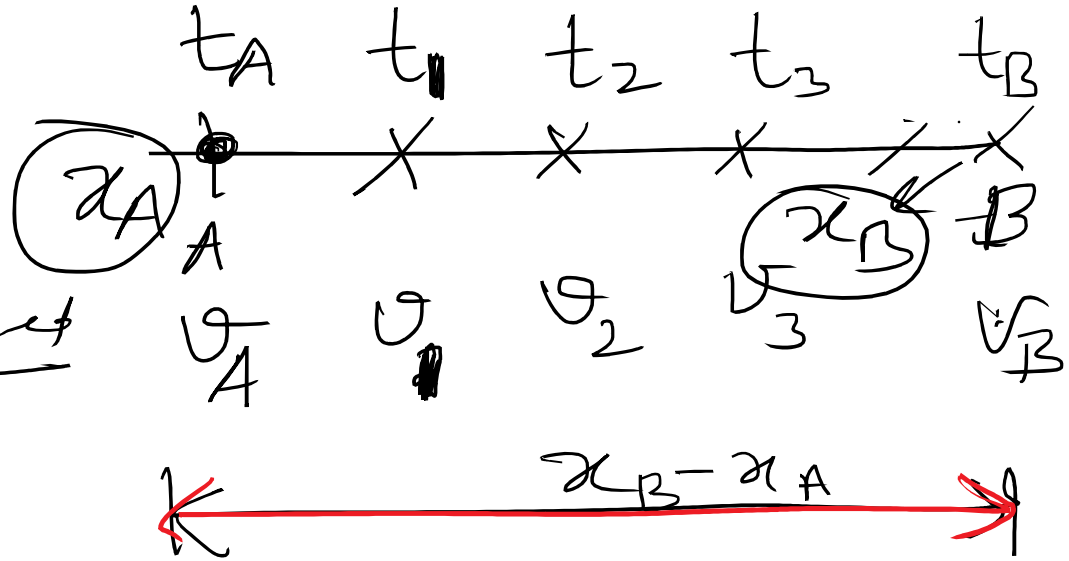
$$\int_0^1 \frac{1}{1+x^2}\,dx = \frac{3}{8}h\left[f_0 + 2\times f_3 + 3(f_1+f_2+f_4+f_5) + f_6\right]$$

$$= \frac{3}{8}\times\frac{1}{6}\left[1 + 2\times 0.8 + 3(0.973+0.9+0.746+0.64) + 0.5\right]$$

$$= \underline{0.78539}$$

Example

Train



$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Total time}}$$

$x(t) \rightarrow t \rightarrow$ time

$$v(t) = \frac{d\,x(t)}{dt} \quad \text{(Relationship)}$$

$$\underbrace{\int_{x_A}^{x_B} dx}_{x_B - x_A} = \text{"}\underbrace{\int_{t_A}^{t_B} v\,dt}\text{"}$$

Integral gives total distance travelled.

Numerical Integration

Example The speed of a train at various time, leaving station A and until station B are given by

	v_0	v_1	v_2	v_3	v_4	v_5	v_6
speed KMPH	0	13	33	39.5	40	40	36
Time (minutes)	0	1/2	1	1.5	2	2.5	3
Time (Hr)	0	1/120	1/60	1/40	1/30	1/24	1/20
	t_0	t_1	t_2	t_3	t_4	t_5	t_6

no of partitions = 6

Simpson's $\frac{1}{3}$rd

$$h = \frac{t_6 - t_0}{6} = \frac{\frac{1}{20} - 0}{6} = \frac{1}{120}$$

$$\text{Distance} = \int_{t=0}^{3} v(t)\,dt$$

$$= \frac{h}{3}\left[v_0 + 4(v_1 + v_3 + v_5) + 2(v_2 + v_4) + v_6\right]$$

$$= \frac{1}{120 \times 3}\left[0 + 4(13 + 39.5 + 40) + 2(33 + 40) + 36\right]$$

$$= \underline{\underline{1.53 \text{ KM}}}$$

Try using $\frac{3}{8}$th !!